

High-performance, Powerful, Programmable



VG710 4G Vehicle Gateway

· LTE CAT6/CAT4

· Wi-Fi 5

· Telematics

· Programmable Edge

1. Product Overview

VG710 4G gateway provides reliable, secure, and programmable in-vehicle connectivity for logistics, transport, heavy equipment, and public safety scenarios.

Positioning: Rugged LTE in-vehicle gateway for fleet telemetry and edge intelligence

Key Features:

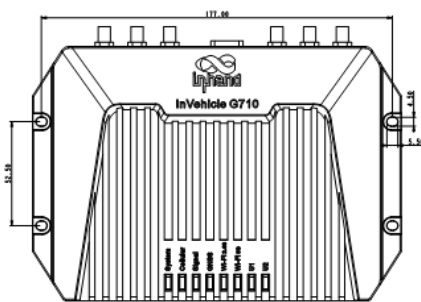
- **Reliable LTE networking:** LTE CAT6/CAT4 with broad global and regional band coverage
- **Vehicle-grade robustness:** IP64 protection, wide input voltage, and industrial design
- **Rich telematics interfaces:** CAN, RS485, DI/DO/AI, and serial ports for vehicle integration
- **Accurate tracking:** GNSS + inertial navigation for stable positioning in complex environments
- **Open edge architecture:** Python/C/C++/Docker with cloud integration APIs

Core Technical Specifications

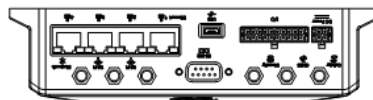
Technical Indicator	Specification
Cellular Network	LTE CAT6/CAT4, supporting global and regional band variants
Positioning Capability	GNSS (GPS/GLONASS/Galileo/Beidou) + inertial DR
Edge Computing	Supports Python, C/C++, and Docker application development and deployment
Cloud & IoT Integration	Supports MQTT, DDS, AMQP, REST, and CoAP; compatible with Azure and third-party platforms
VPN	Supports IPSec/OpenVPN/L2TP/GRE

Technical Indicator	Specification
Wireless Access	Dual-band Wi-Fi 5 (AP/Client) + Bluetooth 4.1
Dimensions	188.1 x 104.5 x 48.8 mm
Weight	775 g
Vehicle Interfaces	4 x Gigabit Ethernet, CAN, RS485, DI/DO/AI, RS232, USB 2.0
Input Voltage Range	9-36 V DC (configurable to 7-36 V DC)
Operating Temperature	-30 C to +70 C
Protection Rating	IP64

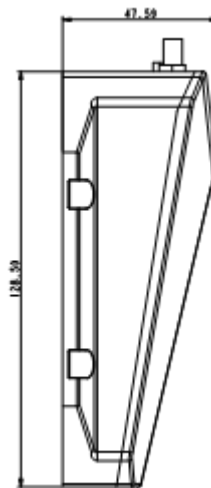
2. Product Dimensions



Top View



Interface Side View



Height Side View

Notes:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and for reference only.
3. Drawings **must not be used for manufacturing**.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category / Parameter	Specification
Performance Metrics	
CPU	ARM Cortex A7
Main Frequency	717 MHz
RAM	1 GB / 512 MB DDR3 (model dependent)
Storage	8 GB eMMC
Connectivity	
Cellular	LTE CAT6 / CAT4
Ethernet	4 × 10/100/1000 Mbps RJ45
Serial Port	RS232 (DB-9)
USB Port	USB 2.0 Micro-B (up to 480 Mbps)
MicroSD	Up to 32 GB, 20 MB/s
Bluetooth	Bluetooth 4.1
Antenna	SMA-K: Cellular/GNSS/Diversity; RP-SMA-K: 2 × Wi-Fi + BLE
Satellite Navigation	
GNSS Receiver	GPS, GLONASS, Galileo, Beidou
Built-in Sensor	Accelerometer + gyroscope, supports DR
Position Accuracy	2.5 m CEP
Tracking Sensitivity	-160 dBm
Location Update Rate	Up to 30 Hz

Category / Parameter	Specification
Wi-Fi	
Frequency	2.4 / 5 GHz dual-band
Protocol	Wi-Fi 5
Max Output	2.4G: 17 dBm; 5G: 17 dBm
Working Mode	AP / Client
Automotive Interfaces	
Diagnostics Interface	CAN bus
DI/DO/AI	6 × DI, 4 × DO, 6 × AI
RS485	RS485 serial (A+, B-, GND)
Other	1-WIRE (driver ID / temperature)
Power	
Input Voltage	9–36 V DC (configurable to 7–36 V DC)
Pin Definition	V+, V-, ignition signal, NC (4 pins)
Protection	Built-in voltage transient protection with delayed ignition induction
Standby Power	0.006 W
Operating Power	12.00 W average
Peak Power	18.20 W peak
Mechanical & Environment	
Installation	Wall-mounting
Protection Rating	IP64

Category / Parameter	Specification
Cooling	Radiation cooling
Housing	Die-cast aluminum
Dimensions (W × D × H)	188.1 × 104.5 × 48.8 mm
Weight	775 g
RTC	Supported
SIM	Dual SIM, 2FF
Operating Temp.	-30 °C ~ +70 °C
Storage Temp.	-40 °C ~ +85 °C
Humidity	95% RH @ 60 °C
Standards & Certifications	
Vehicle Standard	ECE-R10, R118
Rail Standard	EN50155, EN50121, EN61373
Fire Prevention	EN45545-2:2020
Certification	CE, E-Mark, ITxPT, FCC, IC, PTCRB, RoHS, VZW, AT&T, TMO
Warranty	3 years

4. Software Specifications

Category / Parameter	Specification
Network Features	
Network Access	APN, VPDN
LAN Protocol	ARP, Ethernet
Authentication	CHAP/PAP/MS-CHAP/MS-CHAP V2
IP Application	IPv6, Ping, Traceroute, DHCP server/relay/client, DNS relay, DDNS, Telnet, SSH, HTTP, HTTPS, TFTP, FTP, SFTP, Portal
IP Routing	Static routing, RIP, OSPF, BGP, IGMP Proxy
Security	
Firewall	SPI, DoS defense, multicast/Ping filter, ACL, NAT/PAT/DMZ/port mapping/virtual server
User Level	Administrator / read-only
AAA	Local authentication, Radius, Tacacs+, LDAP
CA Certificate	PEM, PKCS12, SCEP
VPN	IPSec VPN, L2TP, GRE, OpenVPN, CA
Reliability	
Backup	Floating routing, VRRP, interface backup
Link Detection	Heartbeat detection, auto redial
Watchdog	Self-detection and auto-repair
Offline Storage	Built-in cache when network unavailable

Category / Parameter	Specification
Ports & WLAN	
VLAN Partition	Supported
Port Mirroring	Supported
WLAN Protocol	IEEE 802.11 b/g/n/a/ac
WLAN Security	Shared key, WPA/WPA2, WEP/TKIP/AES
Configuration	Local/remote HTTP, HTTPS, Telnet, SSH
Upgrade	Local/remote WEB, DM, TFTP, FTP, SFTP server
Diagnostics	Ping, Traceroute, Sniffer
Edge Computing	
Edge Platform	Integrated network-computing-storage-application edge platform
Programmable	Python, C/C++, Docker
SDK	Python 3 SDK, Docker SDK, Azure IoT Edge SDK
IDE	Visual Studio Code
IoT Architecture	MQTT, DDS, AMQP, XMPP, JMS, REST, CoAP
3rd Party Cloud	MS Azure, SmartFleet and APIs for third-party platforms
Docker Images	Node-RED, Ubuntu, Docker for ARM 32, etc.
Application Services	
Fleet Cloud	InHand SmartFleet for route planning, tracking, geofencing, and batch operations
Vehicle Telemetry	Rich interfaces for telemetry and asset tracking devices

Category / Parameter	Specification
Event Alarm	Customizable alarms for DI, network, service, power, temperature, voltage
Message Push	SMS, Email, App, digital output

5. Ordering Information

Model Rule

Model code: VG710-<L/NA>-<WMNN>

<WMNN>: Cellular Type & Module

Product Models

Model	Cellular Type	UE Category	RAM	Region
VG710-L-FQ09	LTE-FDD/LTE-TDD/WCDMA (global bands)	LTE CAT6	1 GB	Global
VG710-L-FQ78	LTE-FDD/TDD + WCDMA + GSM/EDGE	LTE CAT4	1 GB	Latin America, Australia, New Zealand
VG710-FQ09	LTE-FDD/LTE-TDD/WCDMA (global bands)	LTE CAT6	512 MB	Global

VG710-FQ78	LTE-FDD/TDD + WCDMA + GSM/EDGE	LTE CAT4	512 MB	Latin America, Australia, New Zealand
VG710-EN00	NONE	NONE	512 MB	Global

Antenna Options

Antenna	Order Code	Specifications
LTE 4G Antenna	AANT09002 5	LTE/GSM/CDMA multi-band antenna, SMA-J1.5 connector, RG-174, 2000±20 mm
GNSS Antenna	AANT0400 05	GPS/GALILEO/GLONASS GNSS antenna, dimensions 55.6 × 50.5 mm
GNSS Antenna	AANT0400 06	GPS/GALILEO/GLONASS GNSS antenna, dimensions 50 × 38.5 mm
Wi-Fi Antenna (Rubber Ducky)	AANT06001 6	2400–2500 MHz / 4900–5850 MHz, peak gain 5±0.5 dBi
Wi-Fi Antenna (Adhesive)	AANT06001 8	2400–2500 MHz / 4900–5850 MHz, peak gain ≤3 dBi, 2000±20 mm
Bluetooth Antenna (Rubber Ducky)	AANT06001 7	2.4 GHz, peak gain ≤2 dBi

Cable Options

Cable	Order Code	Specifications
Power Cable	SCAB00021 6	4PIN to open-end cable for field engineering; indoor test requires separate power adapter
20 PIN Extension Cord	SCAB00021 9	20PIN extension cable for field engineering and testing
OBD-II Power Cable	SCAB00023 5	20PIN + 4PIN power + OBD-II male + I/O open end + ignition wire
J1939 9PIN Power Cable	SCAB00023 4	20PIN + 4PIN power + J1939 9PIN female + I/O open end + ignition wire
J1939 6PIN Power Cable	SCAB00023 3	20PIN + 4PIN power + J1939 6PIN female + I/O open end + ignition wire
20 PIN to OBD-II	SCAB00021 5	A/B/C/D end cable set for field projects and testing

6. Contact Us

- Website: [InHand Networks](https://www.inhandnetworks.com)
- Copyright: © InHand Networks. All rights reserved.

7. 20PIN Definition

PIN	Definition	PIN	Definition
1	-485	11	485

2	CANL	12	CANH
3	1-Wire	13	GND
4	DO4	14	DO3
5	DO2	15	DO1
6	GND	16	GND
7*	AI6/DI6	17*	AI5/DI5
8	AI4/DI4	18	AI3/DI3
9	AI2/DI2	19	AI1/DI1
10	GND	20	GND

Note:

- 7* : AI6/DI6/FWD
- 17* : AI5/DI5/WHEELTICK

Power Pin Definition (4 pins)

- V+ : Power positive
- V- : Power negative
- Ignition Signal : ACC/IGN input
- NC : Reserved (not connected)